

A Correlated Base–Tail Viability Tube

Signed Support Cocycles beyond Unsigned Projective Products

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Abstract

The directional candidate is the product

$$S(y, a) = a(1 - \sqrt{y})_+^4,$$

where y is a controlled upper for the squared relative tail and $a \in (0, 1]$ is a normalized Gram base. Previous layers bounded the two factors separately. We instead study the correlated support tube

$$\mathcal{K}_\beta = \{(y, a) : S(y, a) \geq \beta\}.$$

If a transition satisfies $y' \leq F(y)$ and $a' \geq ra$, then its sharp tube multiplier is

$$\kappa(y) = r \frac{(1 - \sqrt{F(y)})_+^4}{(1 - \sqrt{y})^4}, \quad S(y', a') \geq \kappa(y)S(y, a).$$

A positive support floor follows whenever the partial sums of $\log \kappa_n$ are uniformly bounded below. This signed-cocycle condition allows losses followed by recoveries and is therefore more flexible than summability of the unsigned projective distances in RH-146. It is sharp for the stated transition information: if the partial sums diverge to $-\infty$, the equality trajectory has zero support liminf.

The outward RH-138 archive contains 328 positive states and 28/30 complete chains in the common tube $\mathcal{K}_{10^{-10}}$. The delayed clean suffix from $\sigma = 0.04$ contains 18/18 chains in a common tube above 1.5×10^{-10} . Local support multipliers include 98 recoveries above one and 194 losses below one. Direct statewise reanchoring improves the RH-146 projective product by between 1.35×10^{16} and 4.44×10^{48} . These are finite correlated certificates; no all-level support cocycle or cross-scale reset is proved.

1 Why a joint tube

Write

$$\phi(y) = (1 - \sqrt{y})_+^4, \quad S(y, a) = a\phi(y).$$

RH-139 uses the sufficient conditions $\limsup y_n < 1$ and $\liminf a_n > 0$. RH-146 gives a sharp projective recurrence for a_n , but its unsigned product loses 16–49 orders of magnitude on the finite chains. The loss is structural: every step is charged its worst possible shape distortion even if a later transition repairs it.

The superlevel set $\mathcal{K}_\beta$ keeps the two factors at their common time index. It is not a logically weaker asymptotic goal. Because $0 < a \leq 1$ and $0 \leq \phi \leq 1$,

$$S(y, a) \geq \beta \implies a \geq \beta, \quad y \leq (1 - \beta^{1/4})^2 < 1. \quad (1)$$

Conversely, $y \leq \bar{y} < 1$ and $a \geq a_* > 0$ imply $S(y, a) \geq a_* \phi(\bar{y})$. Thus a positive eventual joint tube is equivalent in qualitative force to the two RH-139 packets, while permitting a proof to preserve levelwise correlation.

2 Sharp tube transport

Suppose an available control gives a monotone tail upper F and a lower base ratio $r > 0$:

$$y' \leq F(y), \quad a' \geq ra. \quad (2)$$

The base ratio may come from a direct Gram comparison, an interval enclosure, or a block estimate. The theorem itself is independent of its source.

Theorem 1 (Sharp correlated multiplier). *For every source state with $0 \leq y < 1$, transition (2) implies*

$$S(y', a') \geq \kappa(y) S(y, a), \quad \kappa(y) = r \frac{\phi(F(y))}{\phi(y)}. \quad (3)$$

This multiplier is sharp given only (2).

Proof. The function ϕ is decreasing. Hence

$$S(y', a') = a' \phi(y') \geq ra \phi(F(y)) = r \frac{\phi(F(y))}{\phi(y)} S(y, a).$$

Equality occurs when $y' = F(y)$ and $a' = ra$, so no larger universal multiplier follows from the two scalar bounds alone. $\square$

Corollary 2 (One-step tube invariance). *On a source-tail domain $I \subset [0, 1)$, the information (2) preserves every tube $\mathcal{K}_\beta$ if*

$$\inf_{y \in I} \kappa(y) \geq 1.$$

For the stated information set this criterion is exact: if it fails at y_0 , the extremal boundary state $a = \beta/\phi(y_0)$ exits $\mathcal{K}_\beta$ under equality in (2).

If $F(y) \geq 1$, the target factor vanishes and no positive tube can pass that state. The superunit birth obstruction of RH-144–RH-145 is therefore recovered automatically in the joint geometry.

3 Signed support cocycles

One-step invariance is stronger than necessary. A lossy step can be crossed if later recoveries prevent the cumulative product from approaching zero.

Theorem 3 (Lower-bounded logarithmic cocycle). *Suppose a controlled trajectory obeys*

$$S_{n+1} \geq \kappa_n S_n, \quad \kappa_n > 0,$$

for $n \geq N$. If there is $C < \infty$ such that

$$\sum_{j=N}^{n-1} \log \kappa_j \geq -C \quad \text{for every } n > N, \quad (4)$$

then

$$\inf_{n \geq N} S_n \geq S_N e^{-C} > 0.$$

Consequently both conclusions in (1) hold eventually.

Proof. Iteration gives $S_n \geq S_N \exp(\sum_{j=N}^{n-1} \log \kappa_j)$, and (4) gives the stated floor. $\square$

The condition keeps signs. Infinitely many $\kappa_n < 1$ are harmless when they are offset by sufficiently strong $\kappa_n > 1$ steps. This differs from the RH-146 sufficient condition $\sum d_n < \infty$, which charges every projective movement positively.

Proposition 4 (Sharp cocycle obstruction). *If the partial logarithmic sums are unbounded below, no positive universal support floor follows from the multiplier sequence alone.*

Proof. Choose the admissible equality trajectory $S_{n+1} = \kappa_n S_n$. Along a subsequence where the partial sums tend to $-\infty$, this trajectory tends to zero. The constant sequence $\kappa_n = 1/2$ is the simplest witness. $\square$

For controlled families, one may select c_n and use

$$\kappa_{n,c_n}(y_n) = r_{n,c_n} \frac{\phi(F_{n,c_n}(y_n))}{\phi(y_n)}.$$

This is a viability problem on a multiplicative cocycle [Aubin, 1991, Strogatz, 2018]. A block theorem needs a uniform bound on the largest within-block drawdown and a lower bound on all partial sums of block endpoint gains. It does not require every local multiplier to exceed one.

4 Outward finite audit

We combine the independently outward-guarded tail upper and normalized-base lower already archived in RH-138. At each target state we recompute $S = a\phi(y)$; all 330 recomputations agree with the archived support lower. The two superunit states have $S = 0$. Every other state exceeds 1.0365×10^{-10} , so 28 complete chains lie in $\mathcal{K}_{10^{-10}}$.

tube level β	complete chains in $\mathcal{K}_\beta$	interpretation
10^{-10}	28/30	every finite viable chain
10^{-8}	21/30	excludes weak coarse directions
10^{-6}	15/30	retains intermediate/fine packets
10^{-4}	12/30	exactly the two finest anchors

Table 1: Complete-chain correlated tube levels.

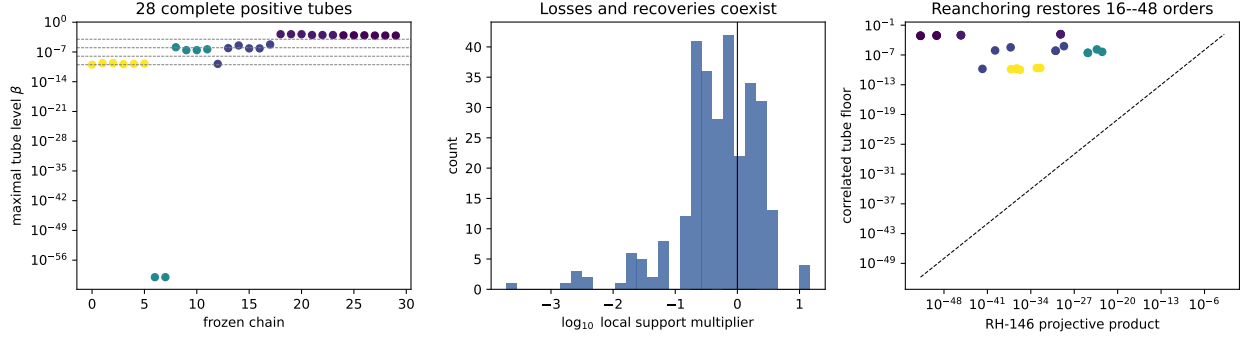


Figure 1: Finite tube levels, signed local support multipliers, and recovery relative to the unsigned RH-146 projective product.

The RH-145 delayed suffix beginning at $\sigma = 0.04$ has 18/18 complete positive chains and common tube level 1.55426×10^{-10} . This is a finite cofinal-anchor statement, not a proof that future anchors enter the same tube.

Among consecutive positive archived states there are 292 local support ratios. Their range is

$$1.9078 \times 10^{-4} \leq S_{n+1}/S_n \leq 14.7601,$$

with median 0.6011. Ninety-eight ratios are recoveries above one and 194 are losses below one. Thus per-step invariance is visibly false, while the finite chains still retain a positive common floor. This is exactly the regime addressed by Theorem 3.

The maximum gain from pairing the worst tail and worst base at their actual common states, rather than as one rectangular chain bound, is only 1.018. The major improvement comes from repeated direct reanchoring: relative to the RH-146 terminal projective products, the correlated tube floors are larger by 1.35×10^{16} to 4.44×10^{48} , with median gain 3.46×10^{26} . An all-level proof must reproduce this reanchoring or signed cancellation analytically; finite direct eigenvalue evaluation cannot substitute for it.

5 Consequence and claim boundary

RH-147 replaces two independently propagated worst-case scalars by one exact support geometry. The new all-level target is:

$$\inf_{n \geq N} \sum_{j=N}^{n-1} \log \left[r_j \frac{\phi(F_j(y_j))}{\phi(y_j)} \right] > -\infty.$$

Together with validated source assembly, this single condition yields an eventual positive directional support floor. It allows isolated losses, recoveries, blocks, and delayed start.

The present paper proves the scalar transport and sharp obstruction and audits a finite outward tube. It does not construct all-level F_n and r_n on one interval source path, prove a cross-scale reset, verify a uniform support cocycle, close Stage A, construct a Hilbert–Polya operator, identify zeta zeros, or prove the Riemann Hypothesis.

References

Jean-Pierre Aubin. *Viability Theory*. Birkhauser, 1991.

Steven H. Strogatz. *Nonlinear Dynamics and Chaos*. CRC Press, second edition, 2018.